

No. of Functional Features Included in the Solution

Date	15 July 2026
Team / Submission ID	
Project Name	Emotion Detection & Learning Support Engine
Submitted By	Venu Gopal (Sole Team Member / Team Lead)
Maximum Marks	5 Marks

Functional Features Overview

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Study Problem Input	Field selector + free-text problem description with quick-example buttons.	app.py (UI)	Done	High
2	BiLSTM Emotion Prediction	Predicts primary emotion + confidence + full probability distribution.	src/bilstm_predict.py	Done	High
3	BERT Emotion Prediction	Second, transformer-based emotion prediction for comparison.	src/bert_predict.py	Done	High
4	Mixed Emotion Detection	Surfaces secondary emotions above a 0.15 probability threshold.	src/mixed_emotion.py , src/utils.py	Done	Medium
5	Keyword-Based Boosting	Nudges probabilities using field-specific emotion keyword lists.	src/keyword_enhancer.py	Done	Medium
6	Gemini AI Response	Generates an empathetic, field-specific learning response via Gemini 2.5 Flash.	src/gemini_helper.py	Done	High
7	Rule-Based Fallback	Serves a predefined response when the Gemini API is unavailable.	src/preprocessing.py (EMOTION_RESPONSES)	Done	High
8	Interaction Logging & Analytics Dashboard	Logs every session to CSV and visualises trends across emotions, fields, and confidence over time.	src/logger.py, app.py (Plotly tabs)	Done	Medium

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5 (input, BiLSTM prediction, BERT prediction, AI response, fallback response)
Additional / Nice-to-Have Features	3 (mixed emotion detection, keyword boosting, analytics dashboard)
Features Tested & Verified	8 (manually verified via local Streamlit runs)

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Study problem input form, analytics dashboard (Emotions / Fields / Summary tabs)
2	Backend / Logic	4	Preprocessing pipeline, mixed emotion detection, keyword boosting, prediction schema
3	Database / Storage	1	CSV-based interaction logging (emotion_response_examples.csv)
4	API / Integration	1	Gemini 2.5 Flash generative AI integration
5	Security / Authentication	0 — not implemented (single-user local prototype, no login)	n/a